# Clype: how the agent works, start to finish.

*Briefing for building the HackTheBurgh 2026 presentation. Written 8 Nov 2026. Figures come from the live dashboard at the time of writing.*

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## 1. One-line pitch

**Clype is a browser agent that clicks "Reject all" on a website's cookie banner, then records what the site still sends anyway, compares that with what its privacy policy promises, has an AI legal reviewer check it against UK law, and drafts (or sends) a complaint letter, all with evidence attached.**

It runs three ways: as a **dashboard** (audit one site or a file of 300), as a **Firefox extension** (audits each site you visit while you browse), and as a **cross-site check** (proves a tracker recognises you across different sites).

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## 2. The problem

- Almost every UK website shows a cookie banner. Most people click "Reject all" and assume that's the end of it.
- Often it isn't: trackers keep loading, cookies get set before you choose, IDs are passed to ad companies, and the privacy policy doesn't name who receives your data.
- Checking this by hand needs developer tools, legal knowledge and hours per site. Regulators (the ICO) can't check every site; individuals can't either.

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## 3. What Clype does about it

For any website, Clype produces:

1. **Evidence**: every network request and cookie, split into *before you chose* and *after you rejected*, matched to the company behind it.
2. **Findings**: plain-English problems with a severity (low / medium / high / critical) and the law each one relates to (UK, with the EU equivalent).
3. **An AI legal review** that must cite the numbered evidence for every claim, and whose quotes from the policy are checked word for word.
4. **A complaint letter** to the company's data protection contact, drafted automatically for serious cases and emailed for critical ones (controlled by a safety switch).
5. **Cost and time** for every audit (tokens, model, seconds), shown on the dashboard.

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## 4. The audit, step by step

This is what happens when one site is audited ( `agent/run_audit.py` ).
Every step is wrapped so that if it fails, the error is recorded and the audit carries on.

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URL —▶ queue —▶ headless Firefox —▶ open site —▶ banner: find Accept (never click)
     —▶ browse 3 pages —▶ read privacy + cookie policy —▶ fresh-browser ID check
     —▶ plain-code checks —▶ AI legal review (+ safeguard) —▶ EU law —▶ letter (+ e
```

### Step 1. Queue

- A URL comes from the dashboard box, an uploaded file (up to 300 URLs), or the Firefox extension.
- It is checked first: local/private addresses are refused, duplicates are dropped by *site* (bbc.co.uk = www.bbc.co.uk), and sites already audited can be skipped (tick box).
- **4 audits run in parallel** (configurable up to 8). Sites you're browsing in the extension take a **fast lane** ahead of waiting batch jobs.

### Step 2. A fresh browser

- A brand-new **headless Firefox** (no cookies, no history) for every audit. Firefox because the extension is for Firefox users, so audits show what they

experience.

- Before the site's own code runs, Clype installs **fingerprinting detectors** (canvas, WebGL GPU model, audio, font probing) that record which script used them.
- **No disguise**: a site that blocks automated browsers is reported as "blocked", never sneaked past. Blocked sites cost no AI calls.

### Step 3. Open the site and watch

- Loads the homepage, waits like a visitor, and records **every request** and **every cookie**, all tagged *before consent*.

### Step 4. The cookie banner

- **Looks for the Accept button but never clicks it**: only to measure whether rejecting is harder than accepting.
- **Checks for pre-ticked toggles** (marketing/analytics switched on by default).
- **Finds and clicks the reject option** using reject-type wording ("Reject all", "Necessary only", "Disable all"…) and only inside a cookie banner, including consent iframes and modern web-component banners (shadow DOM).
- Never clicks "Reject & Pay"/"subscribe" options: those are recorded as **pay-or-consent**.
- If nothing matches, it digs into "Manage preferences" and switches categories off, and as a last resort asks a small AI model to pick the reject button from the visible labels (the AI can only choose among buttons Clype already found).

### Step 5. Browse after rejecting

- Everything from now on is tagged *after reject*. Clype visits up to 3 more pages (one per section, skipping login pages) and scrolls like a person, because many trackers only fire on scroll or navigation.

### Step 6. Read the policies

- Finds the **privacy policy** and **cookie policy** by scoring links (follows "hub" pages one level deeper, reads PDFs too).
- A small AI model extracts the **companies the policy names** and the **data**